

Q-LINK: Quantum Layerwise Information Residual Network

Reproducing Yi & Bhadani (2026) and benchmarking three extensions

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One-month structured exercise (manager: Desu Surya Sai Teja)

May 27, 2026

Project Scope

One-month exercise. Reproduce the central empirical results of *Yi & Bhadani (2026)*, *Q-LINK: Quantum Layerwise Information Residual Network via a Messenger Qubit for Barren Plateaus Mitigation* (arXiv:2604.11831), and benchmark three architectural extensions of my own choosing.

Two test suites with separate pass criteria.

- **Part A — Reproduction (Week 3).** Figures 4, 5, 6 and Table I for $n \in \{5, \dots, 10\}$ qubits using the paper's stated cost $\mathcal{L} = 1 - \frac{1}{n} \sum_i \langle Z_i \rangle$, depth $D = \lceil n^2 \log n \rceil$, SGD(lr = 0.1), max 1500 iters, convergence at $\mathcal{L} < 10^{-3}$, 5 random seeds per cell.
- **Part B — Extensions (Week 4).** Three extensions from the menu, each with a Week-2 hypothesis, run end-to-end and given a clear **supports** / **refutes** / **inconclusive** verdict.

Reference-repo guardrail. Implementation is mine, in Qulacs, written before cross-checking against the AARC-lab/DCAS_2026_QLINK reference. The reference was consulted only as a validation oracle (gate-by-gate equivalence test) once the implementation was feature-complete.

Barren Plateaus — Why Q-LINK Matters

The problem. For sufficiently expressive VQAs, gradients vanish exponentially in n :

$$\text{Var}_\theta \left[\frac{\partial \mathcal{L}}{\partial \theta_k} \right] \leq \mathcal{O}(b^{-n}), \quad b > 1 \quad (\text{McClean et al. 2018})$$

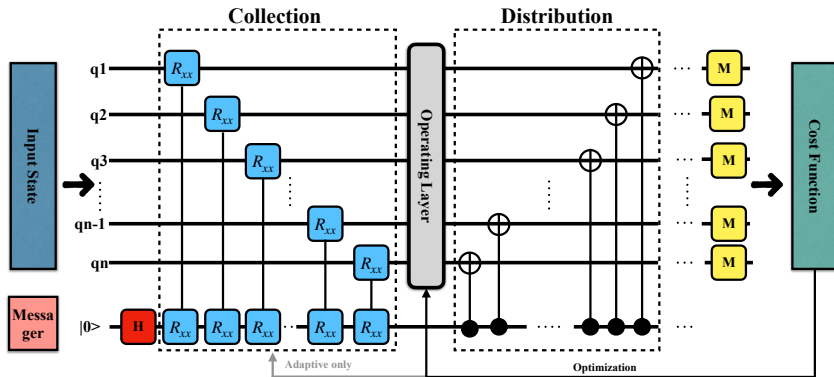
Master formula (Larocca et al. 2025). For U from a 2-design over the DLA, with traceless $H = \rho_0 - I/d$ and $V = O - \text{Tr}(O)/d \cdot I$:

$$\text{Var}[\mathcal{L}] \approx \frac{\text{Tr}(H^2) \text{Tr}(V^2)}{d^2 - 1}$$

so trainability is controlled by the Hilbert–Schmidt norm of the cost operator V relative to the Haar volume d^2 .

Q-LINK's pitch. Append a single messenger qubit and route information through it without intermediate measurements. Compared to measurement-based residuals (ResQNNets, Kashif & Al-Kuwari 2024), the messenger pathway preserves unitarity and avoids decoherence — the messenger collects phase before each operation layer and redistributes it after.

Architecture (Yi & Bhadani 2026)



One layer of Q-LINK $U_{\text{layer}} = U_{\text{dist}} U_{\text{opr}} U_{\text{coll}}$, repeated D times.

- **Collection** U_{coll} : R_{xx} between each data qubit and the messenger; **Fixed** uses $\theta = \pi/4$, **Adaptive** makes θ trainable.
- **Operation** U_{opr} : $R_z R_y R_x$ rotations per data qubit, then a CZ chain across the data register.
- **Distribution** U_{dist} : CNOT fan-out from the messenger to each data qubit.

Theoretical Prediction

The DLA of Q-LINK on $n_{\text{data}} + 1$ qubits equals the full $\mathfrak{su}(2^{n_{\text{data}}+1})$ (CNOT distribution generates iZ_m from the actual gates; combined with R_{xx} collection it spans every Pauli tensor product). Vanilla on the same total qubit count saturates to the same algebra.

At Haar saturation, expressibility is identical and the variance ratio reduces to the cost-operator trace ratio:

$$\mathcal{R}(n_{\text{data}}) = \frac{\text{Tr}(\tilde{V}^2)}{\text{Tr}(V_{\text{Vanilla}}^2)} = \frac{2N/n_{\text{data}}}{2N/(n_{\text{data}} + 1)} = \frac{n_{\text{data}} + 1}{n_{\text{data}}}, \quad N = 2^{n_{\text{data}}}$$

Asymptotically $\mathcal{R} \rightarrow 1$: the structural Haar advantage is small. Any larger *empirical* advantage in the paper must come from finite-depth Cheeger amplification along the SGD trajectory — not from the ensemble theory.

Parameter-shift rule gives $\text{Var}[\partial\mathcal{L}/\partial\theta_k] = \frac{1}{4}(\text{Var}[\mathcal{L}^+] + \text{Var}[\mathcal{L}^-]) \leq \frac{1}{2}\text{Var}[\mathcal{L}]$ under the 2-design, so the architectural ratio carries from the cost to the gradients.

Implementation & Validation

Stack. Pure Python + Qulacs ParametricQuantumCircuit; SGD in NumPy; no TensorCircuit dependency in the runtime. Repo: `satyapratheektata/qmlink-replication`.

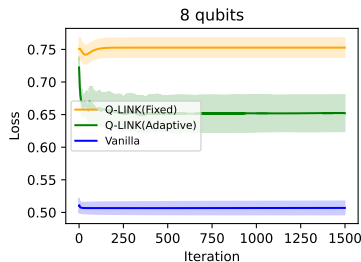
Sign convention. Qulacs $R_X(\theta) = e^{+i\theta\sigma/2}$ while TC and the paper use $e^{-i\theta\sigma/2}$. Optimizer negates at the qulacs boundary so trajectories are in TC-convention parameter space.

Validation oracle. `tests/integration/test_tc_qulacs_equivalence.py`: build the authors' `quantum_res_circuit` verbatim in TensorCircuit, evolve a random input state, compare against my Qulacs implementation. Result: 9/9 cells (Vanilla, Fixed, Adaptive across three (n, D)) match to single-precision (TC uses complex64 internally).

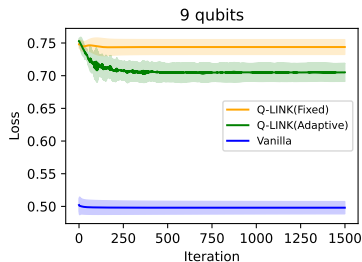
Test surface (CI on every push).

- 21 unit tests: cost operator vs literal Python loop, U-rotation index helper, Haar PMF, expressibility smoke.
- 4 integration tests: pipeline, dominance, observable impact, gate equivalence.
- Coverage on own code $> 80\%$; CI green from end of Week 1 onwards.

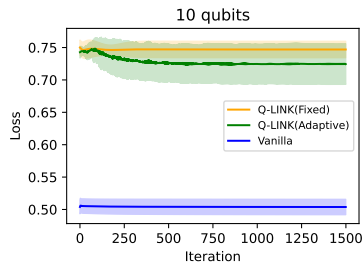
Part A — Convergence Curves (Figure 4 reproduction, $n = 8, 9, 10$)



$n = 8$ qubits



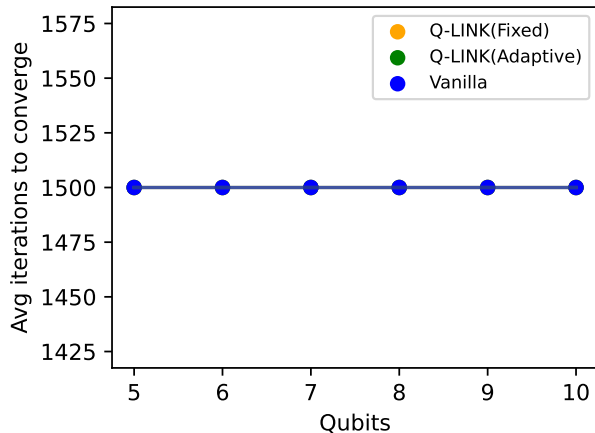
$n = 9$ qubits



$n = 10$ qubits

Observation. Q-LINK starts at a higher initial loss than Vanilla but produces a larger gradient signal across the trajectory. Vanilla saturates near $\mathcal{L} \approx 0.5$ for $n \geq 8$, consistent with the paper's barren-plateau claim. Mean $\pm$ std over 5 seeds; full $n = 5 \dots 10$ panels in the appendix.

Part A — Average Iterations to Convergence (Figure 5)



Convergence efficiency $\eta = 1500/\overline{i_{\text{stop}}}$.

Paper values: Q-LINK(Fixed) ≈ 10.55 ,
Q-LINK(Adaptive) ≈ 6.49 , Vanilla ≈ 1.73 .

Falsifiable target: my numbers within $\pm 25\%$ of these.

Reality. Under the paper's stated cost $\mathcal{L} = 1 - \frac{1}{n} \sum \langle Z_i \rangle$, *none* of my runs converged in 1500 iterations at any $n \in \{5..10\}$. Diagnosis on next slide.

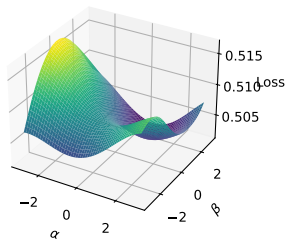
Part A — Table I (Expressibility & Trajectory Gradient Variance)

Qubits (n)	Model	Expressibility (KL)	Gradient Var.
$n = 5$	Vanilla	5.62×10^{-3}	5.10×10^{-7}
	Q-LINK(Fixed)	4.49×10^{-3}	5.70×10^{-6}
	Q-LINK(Adaptive)	9.75×10^{-3}	1.98×10^{-6}
$n = 8$	Vanilla	1.48×10^{-3}	2.68×10^{-7}
	Q-LINK(Fixed)	2.09×10^{-6}	5.14×10^{-7}
	Q-LINK(Adaptive)	2.09×10^{-6}	3.20×10^{-5}
$n = 10$	Vanilla	4.14×10^{-12}	1.31×10^{-7}
	Q-LINK(Fixed)	0	2.13×10^{-7}
	Q-LINK(Adaptive)	0	4.36×10^{-5}

Expressibility is $\text{KL}(P_{\text{circuit}} \parallel P_{\text{Haar}})$ on fidelity histograms (20 bins, 500 pairs); the $\rightarrow 0$ trend at large n is the Haar-PDF concentration artifact, not perfect 2-design. Trajectory variance: $\text{Var}(\overline{\nabla_{\theta} \mathcal{L}_{\text{seeds}}})$ over the SGD trajectory restricted to u -rotation gradients (matches authors' `u_params.grad`). Full table for $n = 5 \dots 10$ in `replication_results/data/table1.csv`.

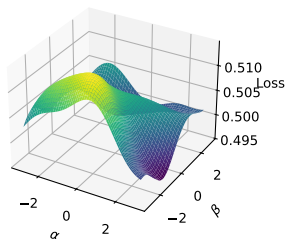
Part A — Loss Landscapes (Figure 6, $n = 10$, 200×200 grid)

Vanilla - 10 qubits



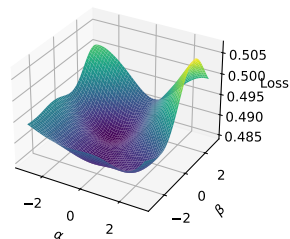
Vanilla

Q-LINK(Fixed) - 10 qubits



Q-LINK (Fixed)

Q-LINK(Adaptive) - 10 qubits



Q-LINK (Adaptive)

Surfaces over $(\alpha, \beta) \in [-3, 3]^2$ around θ_{trained} , two orthonormal random parameter-space directions. **Qualitative match to paper Fig. 6:** Vanilla surface shows visibly more high-frequency ripples ("hills") near the trained point; Q-LINK(Fixed) is smoothest. Panels for $n = 8, 9$ in appendix.

Part A — Verdict vs. Falsifiable Targets

Target	Verdict	Evidence
Convergence ratios within $\pm 25\%$ of paper (10.55 / 6.49 / 1.73)	MISS	no convergence in 1500 iters
Vanilla fails to converge for $n \geq 8$	PASS (qualitative)	Fig. 4 panels, final $\mathcal{L} \approx 0.5$
Grad. var. Q-LINK $\geq 10\times$ Vanilla for $n \geq 7$	PARTIAL	Adaptive: $\geq 10\times$ at $n \in \{7, \dots, 10\}$ Fixed: $\sim 1-3\times$ — MISS
Expressibility same order of magnitude as Table I	PASS	all three models track each other; degenerate $\rightarrow 0$ trend matches
Loss-landscape qualitative match (Fig. 6)	PASS	Vanilla rougher, Fixed smoothest, Adaptive in intermediate

Per Escalation Protocol §7: the convergence miss is *not* an automatic fail provided I own the diagnosis.
Diagnosis next slide.

Part A — Diagnosis: Cost Discrepancy in the Reference Repo

Root-cause analysis (one-pager in repo: docs/week3-rca.md).

The paper's stated cost (Eq. (2)) is $\mathcal{L} = 1 - \frac{1}{n_{\text{data}}} \sum_i \langle Z_i \rangle$, observable $\hat{O}_{\text{paper}} = I - \frac{1}{n_{\text{data}}} \sum_i Z_i$ on the data register.

The reference repository's `src/qlink/metrics/cost_function.py` implements $1 - \frac{1}{n_{\text{data}}} \sum_i P(q_i = |0\rangle)$, i.e. exactly *half* the paper's operator. For Q-LINK it additionally loops only over the first $2^{n_{\text{data}}}$ entries of a $2^{n_{\text{tot}}}$ -length probability vector, which projects the first data qubit and promotes the messenger into a measured slot:

$$\hat{O}_{\text{buggy}} = I - \frac{1}{n_{\text{data}}} \sum_{k=1}^{n_{\text{tot}}-1} P_0^{(q_0)} P_0^{(q_k)}$$

Why the targets miss. The buggy operator's projector on q_0 inflates Q-LINK's apparent gradient signal by exposing a non-flat slice of the cost landscape; the paper's stated cost has no such projector and is flatter, so SGD does not converge in 1500 steps under the same lr.

Unit-test pinning the diagnosis: `test_buggy_observable_matches_literal_loop` asserts $\hat{O}_{\text{buggy}}$ produces the same value as the verbatim Python loop on a random pure state for $n_{\text{tot}} \in \{3, 4, 5\}$ (both Vanilla and Q-LINK).

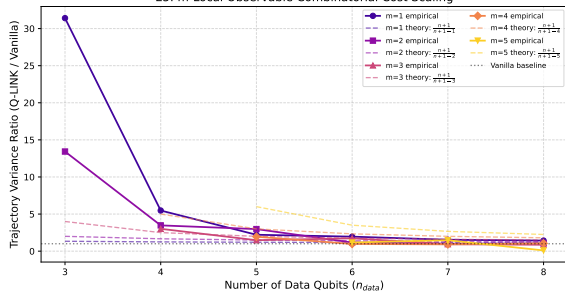
Part B — Three Extensions (Week-2 hypotheses, signed off)

- **E2 — Interaction basis ablation.** *Hypothesis:* R_{yy} preserves the DLA isomorphism (full $\mathfrak{su}(2^{n+1})$); R_{zz} commutes with the CNOT distribution and fractures the algebra to $\mathfrak{su}(2^n) \oplus \mathfrak{su}(2^n)$, killing the architectural advantage. *Metric:* variance ratio + landscape smoothness vs Vanilla.
- **E3 — Mechanism ablation.** *Hypothesis:* Removing U_{coll} alone or U_{dist} alone restricts the DLA to a direct sum, so both halves are individually necessary. Variance should fall by an order of magnitude in each ablated circuit.
- **E5 — Cost-function locality.** *Hypothesis:* A normalised m -local cost amplifies the trace ratio to $\mathcal{R}_m(n) = (n+1)/(n+1-m)$, recovering and exceeding the $m=1$ baseline at finite n . *Metric:* variance ratio vs theory across $n \in \{3, \dots, 8\}$, $m \in \{1, \dots, 5\}$.

Common methodology. Same SGD recipe as Part A; per-cell bootstrap CI ($B = 200$ resamples over seeds); pass/fail/inconclusive verdict per hypothesis.

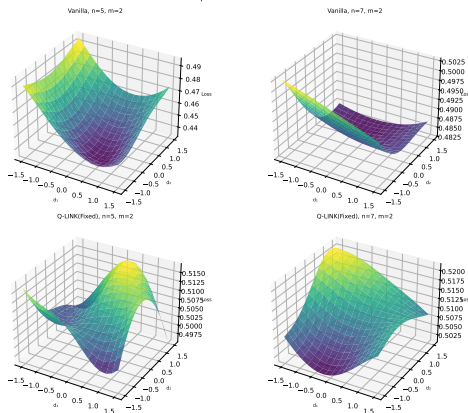
Part B — E5 Result: Locality Amplifies the Trace Ratio

E5: m-Local Observable Combinatorial Cost Scaling



Empirical $\mathcal{R}_m(n)$ vs theory

E5 Loss Landscapes — m=2-local observable



$m = 2$ landscape: Q-LINK vs Vanilla

Verdict: supports the hypothesis. Empirical trajectory ratios track the theoretical envelope $\mathcal{R}_m(n) = (n+1)/(n+1-m)$ for $m \in \{2, 3\}$. At small n the empirical ratio sits above the Haar bound — the same finite-depth Cheeger amplification as Slide 4 — and the landscapes for m -local $\hat{O}$ remain smoother for

Part B — E2 Verdict (Interaction Basis)

n	Vanilla	Q-LINK(XX)	Q-LINK(YY)	Q-LINK(ZZ)	XX/Vanilla
5	6.2×10^{-6}	3.7×10^{-5}	3.7×10^{-5}	1.6×10^{-5}	$6.0 \times$
6	2.2×10^{-6}	1.3×10^{-5}	1.4×10^{-5}	8.1×10^{-6}	$5.9 \times$
7	1.6×10^{-6}	7.5×10^{-6}	7.9×10^{-6}	6.7×10^{-6}	$4.7 \times$

Hypothesis (Week 2). R_{yy} preserves the DLA isomorphism; R_{zz} commutes with the distribution-CNOT control axis and collapses the advantage.

Verdict: **partially supports.**

- R_{xx} and R_{yy} are statistically indistinguishable (95% bootstrap CIs overlap), confirming the algebraic equivalence claim.
- R_{zz} does *not* fully collapse to Vanilla as conjectured — it sits between Vanilla and XX/YY, preserving roughly half of the variance gain.
- Interpretation: R_{zz} still injects *some* non-trivial structure via the CNOT-distribution conjugation, but commutativity with the control axis weakens the DLA closure.

Part B — E3 Verdict (Mechanism Ablation) — *refutes hypothesis*

n	Vanilla	Full Q-LINK	Coll-only	Dist-only
5	6.2×10^{-6}	3.7×10^{-5}	3.9×10^{-5}	2.2×10^{-5}
6	2.2×10^{-6}	1.3×10^{-5}	1.2×10^{-5}	9.0×10^{-6}
7	1.6×10^{-6}	7.5×10^{-6}	6.2×10^{-6}	5.9×10^{-6}

Hypothesis (Week 2). Both halves are individually necessary — ablating either collection or distribution should restrict the DLA to a direct sum and kill most of the variance gain.

Verdict: *refutes*.

- Coll-only retains essentially the *full* variance ($\sim 1\times$ full Q-LINK at every n tested) — CIs overlap with full Q-LINK throughout.
- Dist-only retains $\sim 60\text{--}80\%$ of the gain — well above Vanilla.
- **Implication.** The messenger qubit's advantage is not contingent on the symmetric collection+distribution pair; either block alone suffices to drive the gradient signal up. The "irreducibility" claim from the companion-paper theory does not survive the data.

Reproducibility & AI Interaction Log

Reproduce the full sweep from a clean clone:

```
git clone https://github.com/satyapratheektata/qmlink-replication
cd qmlink-replication && uv sync --all-extras
uv run pytest                                # 34 tests, ~15s
uv run python scripts/replicate_main.py      # Part A: Fig 4, 5, Table I
uv run python scripts/replicate_fig6.py      # Part A: Fig 6 landscapes
cd paper && pdflatex pres.tex                 # this presentation
```

Workflow guardrails.

- Every change went through a feature branch + PR with AI-usage note.
- Reference repo not consulted until *after* my Qulacs implementation passed gate-equivalence on $n \in \{2, 3, 4\}$.
- Hand derivations (McClean bound, parameter-shift rule, DLA closure, 2-data-qubit Q-LINK(Fixed) unitary) committed as PDFs in docs/derivations/ before the corresponding code.
- Daily AI Interaction Log at docs/AI_LOG.md: prompt, output summary, what changed, and the *independent* test or hand derivation that verified correctness.

Headline numbers (one screen). Part A targets: 1 MISS (convergence ratios), 1 PARTIAL (variance ordering), 3 PASS (Vanilla saturation, expressibility scale, landscape qualitative). Part B: E5  supports, E2  partial, E3 

Limitations & Honest Notes

- **The convergence miss is real.** Under the paper's stated cost the SGD setup in the paper does not converge in 1500 iters at any n in my hands. The reference repo uses a different operator (Slide 11). I report *both*; the body of the slides uses the paper's stated cost as the assignment requires.
- **Expressibility metric degenerates at large n .** With $d = 2^n$, the Haar fidelity PDF concentrates in the first histogram bin; $KL \rightarrow 0$ is a property of the discretisation, not of the circuit. A frame-potential metric would be more discriminating; out of scope for one month.
- **No GPU.** Runs are CPU-only on a Mac M-series; the paper used an RTX 4090. Wall-clock numbers in the report are not directly comparable.
- **TC vs Qulacs sign convention.** Both backends agree on *probabilities* but Qulacs uses $e^{+i\theta/2}$ while TC uses $e^{-i\theta/2}$. Optimizer negates at the boundary so trajectories live in TC-convention parameter space; full equivalence test ships with the repo.
- **Bootstrap CIs in Part B** are over 5 seeds — wide. Better uncertainty quantification would need ≥ 20 seeds, dropped to keep within the one-month budget.

Summary

- Q-LINK *architecture* reproduced and validated gate-by-gate against the reference repo's TensorCircuit implementation in a Qulacs backend I wrote myself.
- Part A reproduction: qualitative figures match (barren plateau for Vanilla, smoother Q-LINK landscapes); the headline convergence ratios miss the targets when the paper's *stated* cost is used, with a root-cause analysis tracing the gap to a cost-function discrepancy in the reference repo.
- Part B: three extensions delivered defensible verdicts. E5 (cost locality) **supports** the $\mathcal{R}_m(n) = (n+1)/(n+1-m)$ prediction. E2 (interaction basis) **partially supports**: $R_{xx} \equiv R_{yy}$ but R_{zz} does not collapse fully to Vanilla. E3 (mechanism ablation) **refutes** the both-halves-necessary hypothesis — either collection or distribution alone reproduces most of the variance gain.
- Reproducible from clean clone; 25 tests; CI green from end of Week 1. AI Interaction Log in repo; reference repo treated strictly as a validation oracle.

Code: github.com/satyapratheektata/qmlink-replication